
Notes on exponentially decaying viscosity

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Our problem is to solve

$$F''(z) = 2ik^2\varphi^2 e^{kz}F(z).$$

Define

$$\lambda^2 = 2ik^2\varphi^2, \quad \text{i.e.} \quad \lambda = (1+i)k\varphi,$$

so that

$$F''(z) = \lambda^2 e^{kz}F(z).$$

Introduce the new variable

$$t = \frac{2\lambda}{k} e^{kz/2}.$$

Then

$$\frac{dt}{dz} = \frac{k}{2}t,$$

and therefore

$$\frac{dF}{dz} = \frac{k}{2}t \frac{dF}{dt},$$

$$\frac{d^2F}{dz^2} = \left(\frac{k}{2}\right)^2 \left(t^2 \frac{d^2F}{dt^2} + t \frac{dF}{dt}\right).$$

Since

$$e^{kz} = \left(\frac{tk}{2\lambda}\right)^2,$$

substitution into the differential equation yields

$$t^2 \frac{d^2F}{dt^2} + t \frac{dF}{dt} - t^2 F = 0.$$

This is the modified Bessel equation of order zero,

$$t^2 y'' + t y' - (t^2 + \nu^2) y = 0, \quad \nu = 0.$$

Hence the general solution is

$$F(z) = A I_0(t) + B K_0(t).$$

The boundary condition

$$F(\infty) = 0$$

implies ($A = 0$), since $I_0(t)$ grows exponentially as $t \rightarrow \infty$, whereas $K_0(t)$ decays exponentially. Thus

$$F(z) = B K_0(t).$$

Differentiating gives

$$F'(z) = -\frac{k}{2} B t K_1(t),$$

where we used

$$\frac{dK_0}{dt} = -K_1.$$

Applying the second boundary condition

$$F'(0) = -ifU_g$$

and using

$$t_0 = t|_{z=0} = \frac{2\lambda}{k},$$

one finds

$$-\frac{k}{2} B t_0 K_1(t_0) = -ifU_g,$$

so that

$$B = \frac{ifU_g}{\lambda K_1(t_0)}.$$

Therefore,

$$F(z) = \frac{ifU_g}{\lambda K_1(t_0)} K_0(t),$$

with

$$t = t_0 e^{kz/2}, \quad t_0 = \frac{2\lambda}{k} = 2(1+i)\varphi \quad \implies \quad t = 2(1+i)\varphi e^{kz/2}.$$

Substituting $\lambda = (1+i)k\varphi$ and using

$$\frac{i}{1+i} = \frac{1+i}{2},$$

finally yields

$$F(z) = \frac{(1+i)fU_g}{2k\varphi} \frac{K_0(2(1+i)\varphi e^{kz/2})}{K_1(2(1+i)\varphi)}.$$